APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY (KTU)

CENTRE FOR ENGINEERING RESEARCH AND DEVELOPMENT (CERD)

FORMAT FOR STUDENT PROJECT FUNDING APPLICATION

1. Project Title:

CyberShield AI: A Web-Based Real-Time Cybersecurity Monitoring Dashboard with AI-Assisted Threat Analysis.

2. Broad Area of Research:

Computer Science / Cybersecurity / Artificial Intelligence

3. Details of Principal Investigator (Guide/Faculty):

- Name: Varsha C Mohan

- Designation: Assistant Professor

- Department: Department of Computer Engineering

- Institution: College of Engineering Chengannur, Chengannur (P.O), Alappuzha, Kerala, PIN 689121

- Email & Phone: varshac@ceconline.edu

4. Details of Co-Investigator (if any):

- Name: Geetha S

- Designation & Department: Assistant Professors, Department of Computer Engineering

5. Details of Student Investigators:

1. Abhi Raj (Semester/Branch: 7th Semester, 4th Year B.Tech Computer Engineering)

2. Sreedevi R P (Semester/Branch: 7th Semester, 4th Year B.Tech Computer Engineering)

3. Vishnu V Nair (Semester/Branch: 7th Semester, 4th Year B.Tech Computer Engineering)

6. Project Summary (Abstract):

Existing security operations face a gap between isolated machine-learning threat detectors and practical tools that translate detections into timely, actionable responses. CyberShield AI is a web-based cybersecurity monitoring dashboard designed for small IT teams and SOCs. It aggregates security telemetry, provides on-demand phishing-URL and malware scanning, and leverages the Google Gemini API to generate human-readable explanations for detected threats. The system operates on a React/Vite frontend and an Express.js/TypeScript backend, featuring configurable Discord/Slack webhook notifications for real-time alerting.

7. Objectives of the Project:

- To integrate heuristic and AI-assisted scanning for phishing URLs and malware into a single, centralized dashboard.

- To utilize Large Language Models (Google Gemini API) to generate expert, plain-language explanations for raw security classifications.

- To provide real-time dashboard visualization and severity-categorized automated alerting via webhooks (Discord/Slack) to reduce incident response times.

- To create a lightweight, modular system that can be adopted by small teams without the infrastructure costs of enterprise SOC platforms.

8. Methodology / Work Plan:

The project follows a layered client-server architecture to be completed in 6 months with no external assistance:

- Phase 1 - Frontend Development: Building a responsive, dark-mode-optimized UI using React, Vite, and Tailwind CSS to handle real-time data flows via a 10-second auto-refresh cycle.

- Phase 2 - Backend & API Layer: Implementing an Express.js (Node.js) backend in TypeScript with an in-memory data store to manage metrics, threat analysis, and configurations.

- Phase 3 - AI & Scanner Integration: Integrating the Google Gemini API for expert threat explanations and structuring a VirusTotal API connection for file-hash-based malware scanning.

- Phase 4 - Alerting & Testing: Implementing automated threat categorization and webhook integration for real-time Discord/Slack notifications, followed by system testing.

9. Expected Outcome / Deliverables:

- A fully functional, web-based prototype of the CyberShield AI dashboard.

- An integrated AI Analysis Engine capable of providing natural-language rationale for flagged phishing and malware threats.

- A comprehensive project report and software requirements specification (SRS).

10. Novelty / Innovation:

While extensive research exists on high-accuracy machine learning classifiers for phishing and malware, they are typically evaluated offline and in isolation. The novelty of CyberShield AI lies in directly coupling lightweight threat detection with LLM-generated explanations and real-time, webhook-based alerting. This treats interpretability and immediate actionability as first-class outputs rather than afterthoughts.

11. Budget / Financial Requirements:

SI. No. | Head of Expenditure | Amount (₹)

1 | Equipment / Software: (Cloud Infrastructure, Domain & Hosting) | 4,000

2 | Consumables: (Report Preparation, Printing & Binding) | 2,500

3 | Testing & Analysis: (Google Gemini API token quota & Threat Intel data) | 3,500

4 | Contingencies: | 0

Total | | 10,000

12. Declaration by the Student(s):

We hereby declare that the details furnished above are true and correct. We agree to abide by the rules and regulations of the KTU CERD funding scheme.

Signatures of the Students:

1. _____ (Abhi Raj)

2. _____ (Sreedevi R P)

3. _____ (Vishnu V Nair)

13. Endorsement by the Principal Investigator (Guide):

I recommend this project for financial assistance under the CERD Student Project Scheme. The project is technically feasible and the team is capable of executing it.

Signature: _____

Name: Varsha C Mohan

14. Endorsement by the Head of the Institution (Principal):

Certified that the project investigators are Bonafide students/faculty of this institution. Necessary facilities will be provided to carry out the research work.

Signature & Seal: _____

Name: Prof. (Dr.) Hari. V S (Principal, College of Engineering Chengannur)

Date: